Andrew Thompson

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San Diego, California | U.S. Citizen

EDUCATION

University of California, San Diego *Expected Graduation June, 2027*
B.S. Mechanical Engineering with specialization in Renewable Energy **3.8/4.0 GPA**
- Summer 2025: Study Abroad @ Yonsei, South Korea

EXPERIENCE

Varda Space Industries
Incoming Vehicle Integration & Test Intern *Sept. 2026 - Dec. 2026*

- Selected to support spacecraft assembly, integration, and verification test campaigns for orbital and reentry vehicle systems

Students for the Exploration and Development of Space (SEDS@UCSD)

Lander Lead Propellant Feed Systems Engineer *Jan. 2026 - Present*

- Managed **7 engineers** in design/manufacturing of propellant feed systems of **world's first** collegiate pressure fed **methalox**, 2,000 lbf thrust, 15ft tall, **self-landing** rocket, dedicating **25+ hrs/week**
- Mentored new members on Lander's fluid systems, enabling them to take **immediate ownership** of component design and manufacturing, facilitating **knowledge transfer**
- Spearheaded **communication** between fluids and avionics to synchronize solenoid valve actuation time and sensor data acquisition for flight critical maneuvers
- Coordinated **200psi** vehicle **leak check** with custom test setup, and led following troubleshooting
- Optimized component sourcing (valves, **AN, NPT, Swagelok fittings**) organizing full **BOM**

Lander Propellant Feed Systems Engineer *June 2025 - Jan. 2026*

- Assembled full propellant feed system, prioritizing **reusability** and **affordability**, accounting for extreme **thermal conditions**, **collapse factor**, **boil off**, and pressure budget
- Updated **Solidworks assembly** of rocket, including full fluids system with **300+ unique components**
- Authored and maintained custom **P&ID** with over **30+ unique components**
- Standardized system documentation of **solenoids, pressure transducers, and thermocouples** to facilitate collaboration across structures, avionics and controls
- Developed a custom **MATLAB** script to calculate system head loss, accounting for **major/minor losses**, ensuring propellant **mass flow rates** meet engine requirements

Colossus Fluids Engineer *Jan. 2025 - June 2025*

- Supported fluids manufacturing of Colossus, a **static fire test stand** for liquid bi-propellant rocket engines to give collegiate teams **flexible, safe, and reliable** engine testing rated up to **2700 lbf**
- Manufactured custom components with **SolidWorks**, **3D printing**, and **CNC/manual** machining for structures and electronics integration
- Restored test stand functionality following FOD-initiated failure by redesigning fluid lines and performing precision **cutting, bending, and flaring** of **stainless steel tubes** for full scale, high pressure, hot fire testing applications
- Enhanced safety by designing and implementing **nitrogen-based fire suppression system**

SKILLS

Design & Software: P&ID (Lucid Chart), CAD (SolidWorks, Fusion360), MATLAB, Microsoft Office
Manufacturing & Tools: Soldering, mills, lathes, CNC, 3D printing & prototyping
Soft Skills: Leadership, communication, teamwork, collaboration, personable, organized